

Entity Analysis Report

E-Commerce Order Management Database System — Week 2 Deliverable

1. Introduction

This report continues the E-Commerce Order Management Database System project from the Week 1 Software Requirement Specification (SRS). It identifies the complete set of entities required by the system, lists every attribute for each entity, and specifies the Primary Key (PK), Foreign Key (FK), and column-level constraints — NOT NULL, UNIQUE, CHECK, and DEFAULT — needed to enforce data integrity when the schema is implemented in SQL. Only the nine core entities and attributes fixed in Week 1 are used; no entity or attribute has been added, removed, or renamed.

2. Entity Identification

Nine entities were identified from the Week 1 SRS as required to fully model the business operations of the e-commerce platform.

Entity	Classification / Role
Customer	Master entity for registered users of the platform.
Category	Master entity used to classify products.
Product	Master entity representing an item available for sale.
Supplier	Master entity recording vendors who provide products.
Order	Transaction entity representing a customer purchase.
Order Details	Associative/transaction entity holding the line items of an Order.
Payment	Transaction entity recording payment against an Order.
Shipment	Transaction entity recording delivery of an Order.
Review	Transaction entity recording customer feedback on a Product.

3. Entity Attribute List with Keys and Constraints

The table below lists every attribute for each entity together with a suggested SQL data type and the constraints (PK, FK, NOT NULL, UNIQUE, CHECK, DEFAULT) required to enforce the business rules identified in Week 1.

3.1 Customer

Attribute	Data Type	Constraints
Customer_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Name	VARCHAR(100)	NOT NULL
Email	VARCHAR(100)	NOT NULL, UNIQUE
Mobile_Number	VARCHAR(15)	NOT NULL, UNIQUE, CHECK (length between 10 and 15 digits)
Address	VARCHAR(255)	NOT NULL
Registration_Date	DATE	NOT NULL, DEFAULT CURRENT_DATE

3.2 Category

Attribute	Data Type	Constraints
Category_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Category_Name	VARCHAR(50)	NOT NULL, UNIQUE
Description	VARCHAR(255)	NULL allowed

3.3 Product

Attribute	Data Type	Constraints
Product_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Product_Name	VARCHAR(100)	NOT NULL
Price	DECIMAL(10,2)	NOT NULL, CHECK (Price > 0)
Stock_Quantity	INT	NOT NULL, DEFAULT 0, CHECK (Stock_Quantity >= 0)
Category_ID	INT	FOREIGN KEY REFERENCES Category(Category_ID), NOT NULL

3.4 Supplier

Attribute	Data Type	Constraints
Supplier_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Supplier_Name	VARCHAR(100)	NOT NULL

Attribute	Data Type	Constraints
Contact_Information	VARCHAR(100)	NOT NULL, UNIQUE

Note: The fixed Week 1 attribute list does not include a linking attribute (e.g. Supplier_ID) inside Product. Supplier is therefore modelled as a standalone master entity in the current schema — see Section 5 for further discussion.

3.5 Order

Attribute	Data Type	Constraints
Order_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Customer_ID	INT	FOREIGN KEY REFERENCES Customer(Customer_ID), NOT NULL
Order_Date	DATE	NOT NULL, DEFAULT CURRENT_DATE
Total_Amount	DECIMAL(10,2)	NOT NULL, DEFAULT 0.00, CHECK (Total_Amount >= 0)
Order_Status	VARCHAR(20)	NOT NULL, DEFAULT 'Pending', CHECK (Order_Status IN ('Pending','Confirmed','Shipped','Delivered','Cancelled'))

3.6 Order Details

Attribute	Data Type	Constraints
Order_Detail_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Order_ID	INT	FOREIGN KEY REFERENCES Order(Order_ID), NOT NULL
Product_ID	INT	FOREIGN KEY REFERENCES Product(Product_ID), NOT NULL
Quantity	INT	NOT NULL, CHECK (Quantity > 0)
Unit_Price	DECIMAL(10,2)	NOT NULL, CHECK (Unit_Price > 0)

Note: UNIQUE (Order_ID, Product_ID) is recommended so the same product is not duplicated as two separate lines within one order.

3.7 Payment

Attribute	Data Type	Constraints
Payment_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Order_ID	INT	FOREIGN KEY REFERENCES Order(Order_ID), NOT NULL, UNIQUE
Payment_Method	VARCHAR(30)	NOT NULL, CHECK (Payment_Method IN ('Credit Card','Debit Card','Net Banking','Digital Wallet','Cash on Delivery'))
Payment_Date	DATE	NOT NULL, DEFAULT CURRENT_DATE
Payment_Status	VARCHAR(20)	NOT NULL, DEFAULT 'Pending', CHECK (Payment_Status IN ('Pending','Successful','Failed','Refunded'))

Note: UNIQUE on Order_ID enforces the one-to-one relationship between Order and Payment.

3.8 Shipment

Attribute	Data Type	Constraints
Shipment_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Order_ID	INT	FOREIGN KEY REFERENCES Order(Order_ID), NOT NULL, UNIQUE
Delivery_Address	VARCHAR(255)	NOT NULL
Shipment_Date	DATE	NULL allowed (set once dispatched)
Delivery_Status	VARCHAR(20)	NOT NULL, DEFAULT 'Processing', CHECK (Delivery_Status IN ('Processing','Dispatched','In Transit','Delivered','Returned'))

Note: UNIQUE on Order_ID enforces the one-to-one relationship between Order and Shipment.

3.9 Review

Attribute	Data Type	Constraints
Review_ID	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL
Customer_ID	INT	FOREIGN KEY REFERENCES Customer(Customer_ID), NOT NULL
Product_ID	INT	FOREIGN KEY REFERENCES Product(Product_ID), NOT NULL
Rating	INT	NOT NULL, CHECK (Rating BETWEEN 1 AND 5)
Comments	VARCHAR(500)	NULL allowed

4. Primary Key and Foreign Key Summary

Entity	Primary Key	Foreign Key(s)
Customer	Customer_ID	—
Category	Category_ID	—
Product	Product_ID	Category_ID → Category(Category_ID)
Supplier	Supplier_ID	— (see Section 5 note)
Order	Order_ID	Customer_ID → Customer(Customer_ID)
Order Details	Order_Detail_ID	Order_ID → Order(Order_ID); Product_ID → Product(Product_ID)
Payment	Payment_ID	Order_ID → Order(Order_ID)
Shipment	Shipment_ID	Order_ID → Order(Order_ID)
Review	Review_ID	Customer_ID → Customer(Customer_ID); Product_ID → Product(Product_ID)

5. Notes and Assumptions

- All primary keys are implemented as surrogate, auto-incrementing integer keys, which is standard practice for transaction-heavy e-commerce schemas and keeps foreign key joins efficient.
- CHECK constraints are used to enforce valid value ranges (e.g. Rating between 1 and 5, Price greater than 0) and valid status vocabularies (e.g. Order_Status, Payment_Status, Delivery_Status), preventing invalid data at the database level.
- DEFAULT constraints are applied to date and status columns so that a sensible value is recorded automatically when a row is created, reducing the chance of incomplete records.
- UNIQUE constraints on Payment.Order_ID and Shipment.Order_ID enforce the one-to-one relationships identified in Section 5 of the Entity and Relationship Analysis document.
- Per the fixed Week 1 attribute list, Product does not carry a Supplier_ID attribute, so no foreign key can be defined between Supplier and Product without adding a new attribute — which the project instructions do not permit. Supplier is therefore retained as a standalone master entity in this phase; if a future requirement change allows it, a Supplier_ID foreign key on Product (or a Product_Supplier bridge entity) would resolve this.